RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

IPL CRICKET ANALYTICS DASHBOARD

Using Microsoft Power BI

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ABSTRACT

The IPL Cricket Analytics Dashboard is a data visualization project developed using Microsoft Power BI. It presents cricket-related information in an interactive and easy-to-understand format, helping users explore match statistics, team performance, player contributions, and trends across seasons. The dashboard uses charts, cards, and other visual elements to turn structured match data into useful summaries.

The project workflow includes loading the dataset, inspecting and preparing the data, creating visualizations, and arranging them into a report. The dashboard is intended to support exploratory analysis and comparison rather than predict future match outcomes. Screenshots of the completed dashboard can be inserted in the designated spaces throughout this report.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Cricket generates a large amount of information about teams, players, matches, innings, and results. A dashboard helps organize this information and makes patterns easier to understand. This project uses Microsoft Power BI to create an IPL cricket analytics dashboard that presents important match-related information through visual reports.

1.2 Problem Statement

Raw cricket data is often stored in tables containing many records and fields. Manually reviewing the data makes it difficult to compare teams, understand trends, and identify useful statistics quickly. A visual analytics dashboard can summarize the information and present it in a clear and interactive manner.

1.3 Objectives

- Import and inspect the cricket dataset.
- Prepare data for analysis and visualization.
- Present important cricket statistics using Power BI visuals.
- Compare team and match-related measures through charts.
- Provide an easy-to-understand dashboard for exploratory analysis.

1.4 Scope of the Project

The project focuses on visual analysis of the cricket data available in the dataset used for the Power BI report. The scope is limited to the fields and records included in the source data. The dashboard is not a live score system unless connected to a continuously updated data source.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Microsoft Power BI

Microsoft Power BI is the primary tool used to create the dashboard. It supports data import, transformation, data modeling, and the creation of interactive visualizations such as cards, bar charts, line charts, and pie or donut charts.

2.2 Dataset File

The cricket dataset provides structured records for analysis. Depending on the source file, its columns may include match details, teams, dates, venues, innings information, and results. The dataset is imported into Power BI before visuals are created.

2.3 Power Query and Data Preparation

Power Query can be used to inspect column names, adjust data types, remove unnecessary fields, handle missing values, and prepare the data for reporting. Data preparation improves the consistency and usability of the dashboard.

2.4 Technologies Summary

Technology	**Purpose**
Microsoft Power BI	Creates interactive reports, charts, and dashboard pages.
Power Query	Supports data cleaning and transformation before visualization.
CSV / structured dataset	Stores the cricket match data in rows and columns.
DAX (if used)	Creates calculated measures and values for analysis.

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The dashboard is based on the cricket dataset imported into the supplied Power BI project. The dataset contains the records and fields used to build the report visuals. The exact row count, column count, and date range should be confirmed from the source dataset in Power BI before final submission.

3.2 Important Attributes

Attribute / Field	Description
Match / ID	Identifies a match record, where available.
Season / Date	Represents the season or date associated with a match.
Team fields	Identify the teams participating in a match.
Venue / City	Describe where the match was played, where available.
Innings / Score	Contain score-related information, where available.
Result / Winner	Represent the recorded match outcome.
Player fields	Contain player-related information, where available.

3.3 Dataset Quality Checks

Before building the report, the dataset should be checked for missing values, duplicate records, inconsistent names, and incorrect data types. These checks help reduce errors in the visualizations. Any data quality figures should be reported only after they have been verified in the actual dataset.

CHAPTER 4: METHODOLOGY

4.1 Project Workflow

Dataset Import → Data Inspection → Data Preparation → Visual Creation → Dashboard Design → Interpretation

4.2 Data Inspection and Preparation

1. Open the Power BI project and identify the imported dataset.
2. Inspect the available fields and their data types.
3. Check for missing values, duplicate records, and inconsistent categories.
4. Transform or clean the fields as required using Power Query.
5. Create measures or calculated columns only where needed.

4.3 Dashboard Development

Suitable visualizations are selected according to the type of data and the question being analyzed. Cards can summarize key values, bar charts can compare categories, and line charts can show changes over time. Related visuals are arranged on report pages so users can read the dashboard logically.

4.4 Interpretation

The completed visuals are reviewed to identify visible trends and comparisons. Interpretations should be based on the values shown in the dashboard and should not claim causes or predictions that the dataset cannot establish.

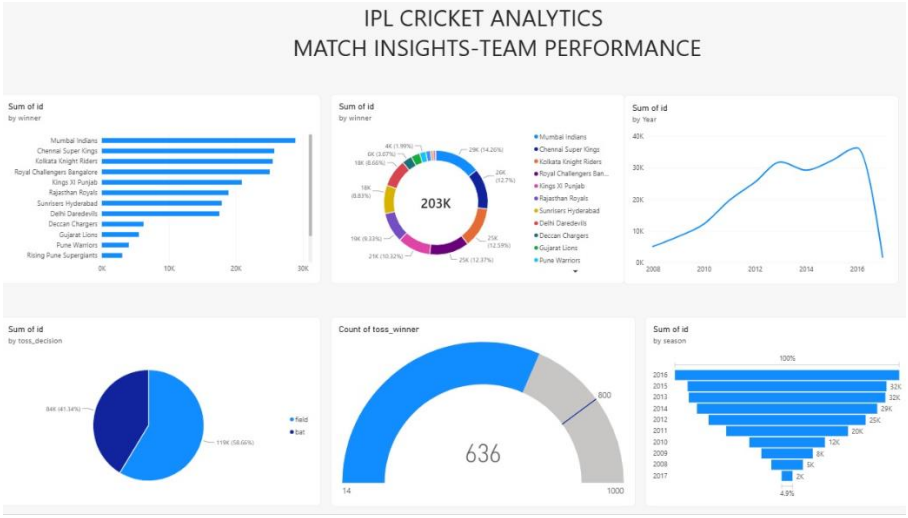
CHAPTER 5: POWER BI DASHBOARD AND VISUALIZATIONS

5.1 Dashboard Overview

The Power BI dashboard brings cricket-related measures together in a visual report. The following spaces are reserved for screenshots from the user's own PBIX report. Replace each placeholder with a clear screenshot showing the relevant page or visual.

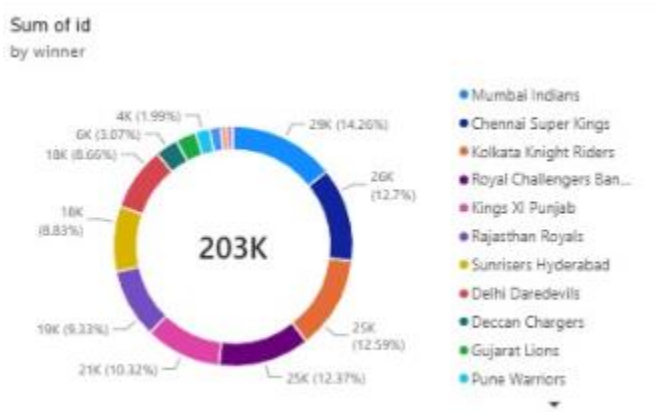
5.2 Dashboard Page / Overview

This section should describe the main dashboard page, including the key performance cards and the main charts displayed.



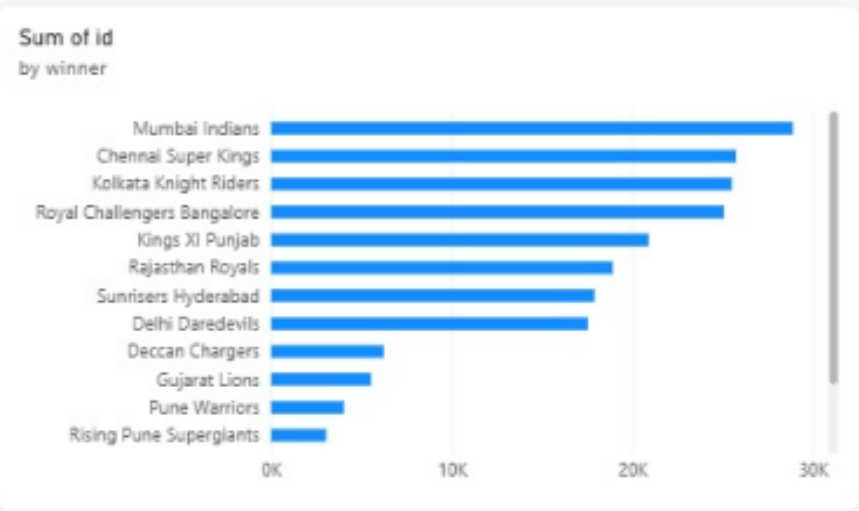
5.3 Team Performance Analysis

Use this section to explain the team comparison visuals available in the report. Mention the leading categories only after checking the values in the actual dashboard.



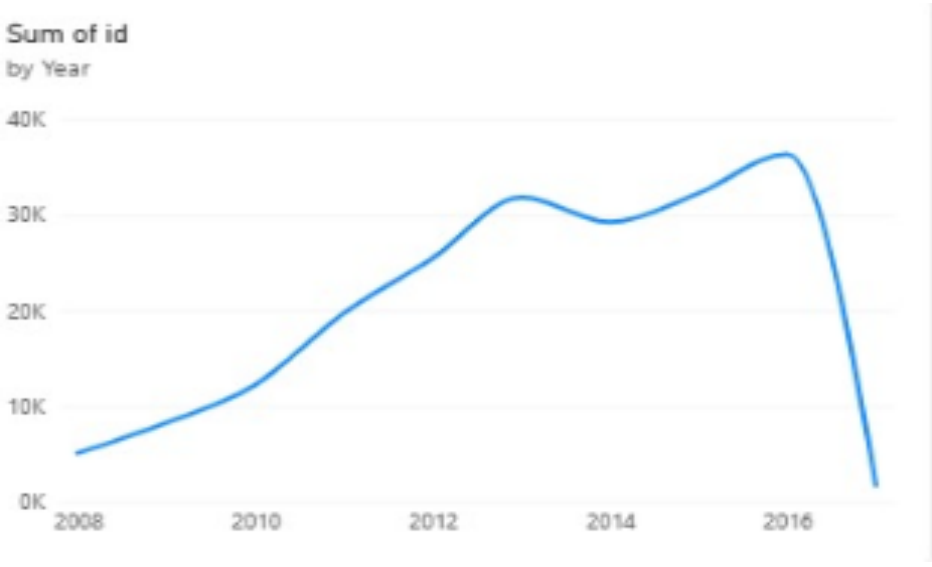
5.4 Match / Season Trend Analysis

If the report includes a season-wise or date-wise chart, use it to describe changes in match activity or the selected cricket statistic over time.



5.5 Additional Visualizations

Add a short explanation of any other visuals included in the project, such as player statistics, score comparisons, venue analysis, or match outcomes. Retain only the sections that match the visuals in the PBIX file.



CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

The Power BI report converts the structured cricket data into visual summaries that are easier to review than raw rows alone. The dashboard can support comparisons across teams, seasons, and other available match attributes. Add specific numerical findings after confirming them in the report.

6.2 Discussion

Visualizations make it easier to identify differences and patterns in the dataset. However, the displayed counts depend on the quality and coverage of the source data. A larger count does not necessarily indicate better performance unless it is considered alongside relevant context, such as matches played or opportunities.

CHAPTER 7: ADVANTAGES, LIMITATIONS AND FUTURE ENHANCEMENTS

7.1 Advantages

- Summarizes cricket data in a clear visual format.
- Makes category and season comparisons easier.
- Reduces the effort needed to inspect raw data manually.
- Supports interactive exploration when filters and slicers are configured.
- Provides practical experience with data preparation and business intelligence.

7.2 Limitations

- The accuracy of the results depends on the source dataset.
- The dashboard can only analyze fields that are available in the dataset.
- Counts alone may not represent performance fairly without context.
- The report is not automatically live unless connected to an updating source.
- Conclusions should not go beyond what the displayed data supports.

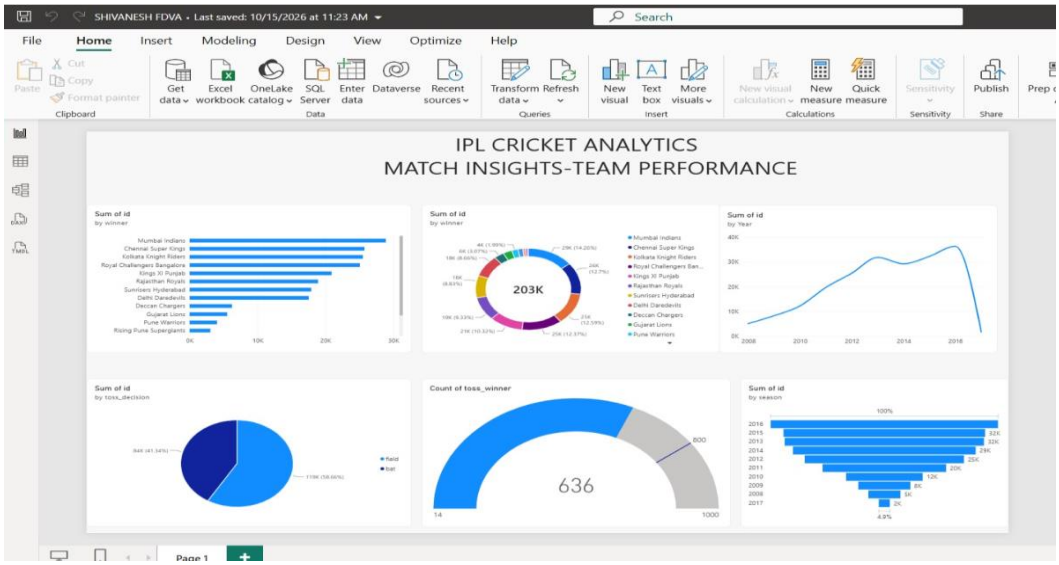
7.3 Future Enhancements

- Add slicers for season, team, venue, and match result where relevant.
- Create additional performance measures, such as percentages or averages, if supported by the data.
- Improve tooltips and labels to make charts easier to interpret.
- Add drill-through pages for detailed team or season analysis.
- Connect to a refreshed data source if up-to-date reporting is required.

CHAPTER 8: DASHBOARD OUTPUT AND CONCLUSION

8.1 Dashboard Output

Insert the final screenshot of the completed Power BI report below. Ensure the screenshot is readable and includes the dashboard title, visual labels, and report page tabs where possible.



8.2 Conclusion

The IPL Cricket Analytics Dashboard demonstrates how Microsoft Power BI can be used to explore and present cricket data through visualizations. The project workflow covers dataset import, inspection, preparation, visual design, and interpretation. The report makes cricket-related information easier to compare and understand.

The dashboard also provides practical experience in data visualization and report development. The findings should be interpreted within the limits of the available dataset, and all numerical claims should be verified against the final Power BI report before submission.